

Chandan Kumar

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EDUCATION

Sardar Vallabhbhai National Institute of Technology

B.Tech in Computer Science & Engineering; CGPA: 8.60/10

Surat, India

2024 – 2028

Bihar School Examination Board

Class XII (Science) — 396/500 (79.2%) | Class X — 443/500 (88.6%)

Bihar, India

2020 – 2022

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, SQL, HTML/CSS

Frameworks: React, Next.js, Node.js, Express.js, FastAPI, Tailwind CSS

Databases & Tools: PostgreSQL (pgvector), MySQL, MongoDB, Prisma, Git, GitHub, VS Code

AI / Computer Vision: YOLO, OpenCV, vector search, OCR

PROJECTS

MindSettler — Mental Wellness Platform | *Next.js, PostgreSQL, Prisma, LangChain*

[Live Demo](#)

- Co-developed a 4-person wellness platform on Next.js 16 and React 19, exposing 95 HTTP handlers across 61 App Router API files over a 23-model PostgreSQL/Prisma schema
- Engineered a Gemini 2.5 Flash assistant with a 2-pass LangChain tool-calling loop, 6-field Zod booking contract, and 2,048-token cap for validated session requests
- Secured 6 route families for 3 roles through NextAuth JWT edge middleware, bcrypt cost-10 credentials, Google OAuth linking, and 10-minute single-use reset tokens
- Implemented 2 payment paths—Razorpay HMAC-SHA256 and PhonePe SHA-256 callback/server verification—driving appointment transitions and 1-month membership expiry
- Built a persistent 5-phase wellness journey with 15 guided activities, 6-state mood check-ins, localStorage progress recovery, and interval-driven breathing exercises

Descriptive CCTV Search | *Python, PostgreSQL, FastAPI, Next.js*

July 2026 – Present

- Architected a surveillance-search stack indexing 41.5K tracklets and 74K multi-view crops from 14 scene-camera streams as 1152-D SigLIP vectors in a 1.7 GB PostgreSQL/pgvector database
- Accelerated fixed-camera ingestion 71% ($0.56\times$ to $0.96\times$ real time) by disabling unnecessary BoT-SORT optical-flow compensation and capping CPU pools, cutting 12-core load average from 25 to 7.7
- Streamed 14.9K live tracklets through a 4-stage bounded pipeline with 8-frame/256-tracklet backpressure, per-tracklet commits, and 2-worker asynchronous CPU OCR off the GPU path
- Benchmarked 4 re-identification encoders on 737 cross-camera queries, retaining FastReID at 47.1% rank-1/63.2% rank-5 after DINOv2 reached 10.9% and Veri-Wild 38.0%
- Shipped 4 search modes and 5 case APIs with immutable audit trails; recorded 61 searches and 7 append-only export receipts for SHA-256/Ed25519-signed packages up to 623 MB

ACHIEVEMENTS & LEADERSHIP

Solved **150+** Data Structures and Algorithms problems across LeetCode and Codeforces, with active contest practice

Clubs: Nexus Club — Representative; ACM Student Chapter — Executive Member; GDG on Campus — Junior Developer